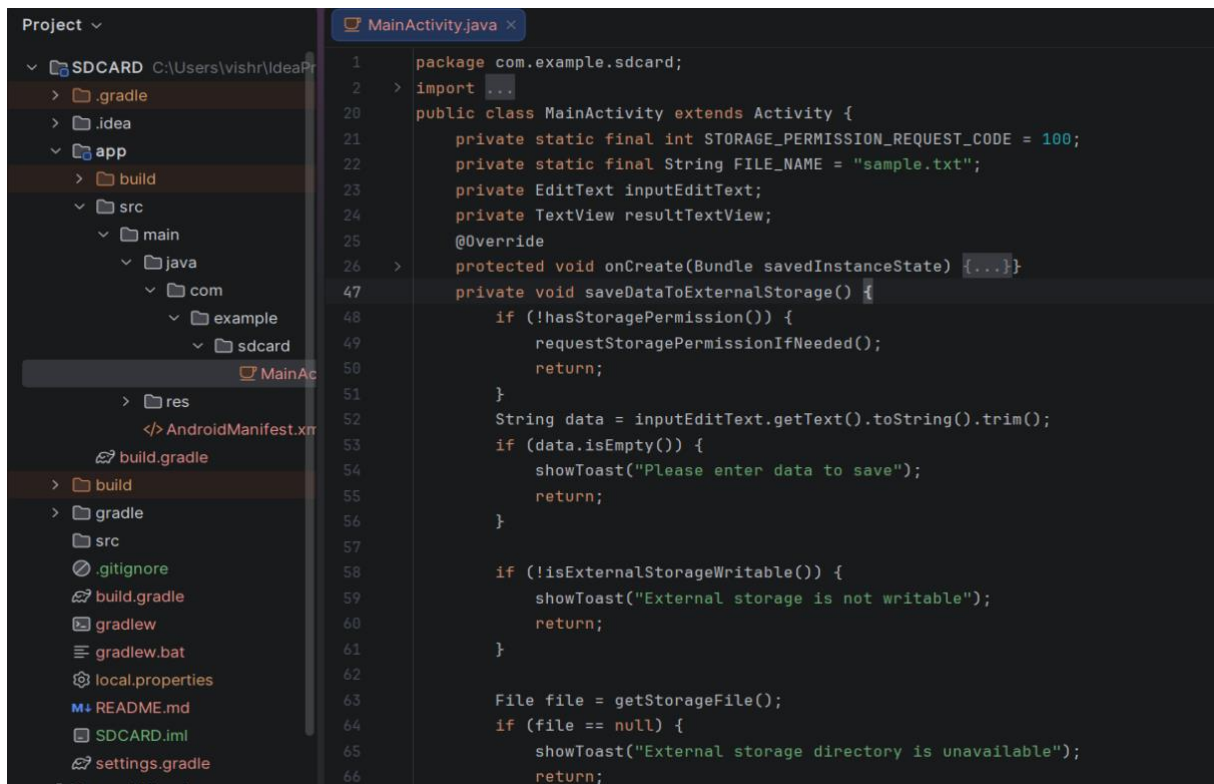


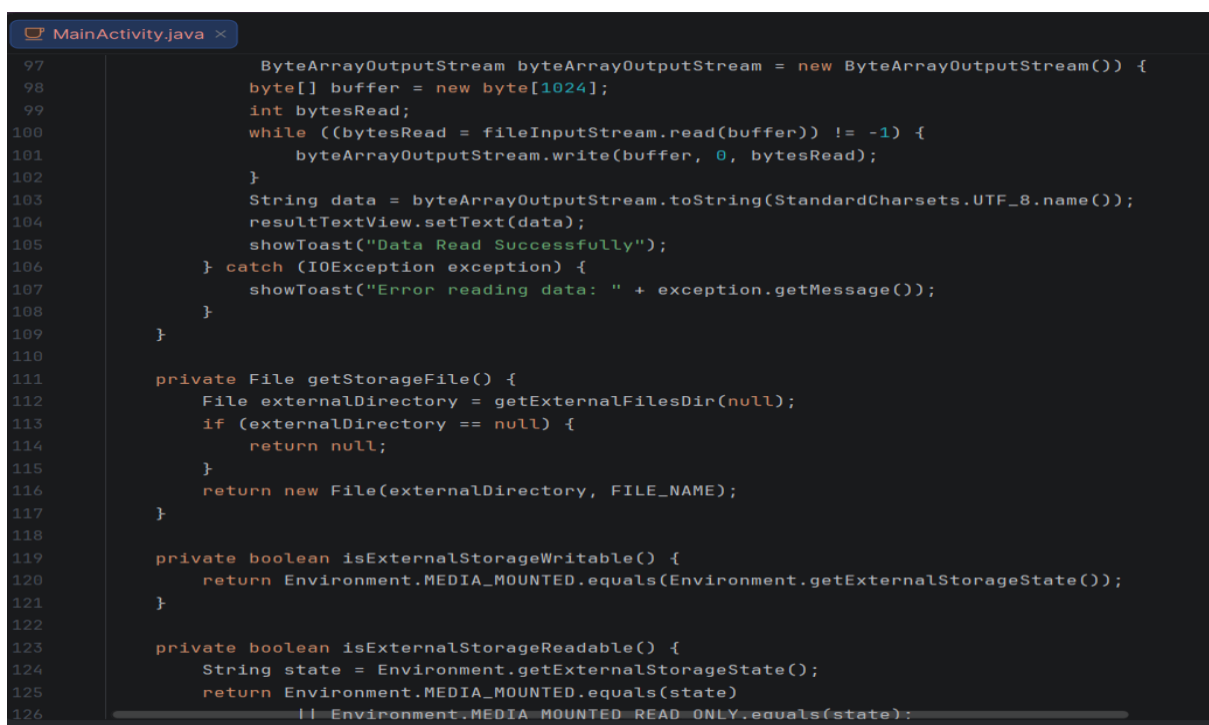
K S VISHRUTHI

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11. Implement an application that writes data to the SD card.



```
1 package com.example.sdcard;
2 import ...
20 public class MainActivity extends Activity {
21     private static final int STORAGE_PERMISSION_REQUEST_CODE = 100;
22     private static final String FILE_NAME = "sample.txt";
23     private EditText inputEditText;
24     private TextView resultTextView;
25     @Override
26     protected void onCreate(Bundle savedInstanceState) {...}
47     private void saveDataToExternalStorage() {
48         if (!hasStoragePermission()) {
49             requestStoragePermissionIfNeeded();
50             return;
51         }
52         String data = inputEditText.getText().toString().trim();
53         if (data.isEmpty()) {
54             showToast("Please enter data to save");
55             return;
56         }
57
58         if (!isExternalStorageWritable()) {
59             showToast("External storage is not writable");
60             return;
61         }
62
63         File file = getStorageFile();
64         if (file == null) {
65             showToast("External storage directory is unavailable");
66             return;
```



```
97         ByteArrayOutputStream byteArrayOutputStream = new ByteArrayOutputStream() {
98             byte[] buffer = new byte[1024];
99             int bytesRead;
100             while ((bytesRead = fileInputStream.read(buffer)) != -1) {
101                 byteArrayOutputStream.write(buffer, 0, bytesRead);
102             }
103             String data = byteArrayOutputStream.toString(StandardCharsets.UTF_8.name());
104             resultTextView.setText(data);
105             showToast("Data Read Successfully");
106         } catch (IOException exception) {
107             showToast("Error reading data: " + exception.getMessage());
108         }
109     }
110
111     private File getStorageFile() {
112         File externalDirectory = getExternalFilesDir(null);
113         if (externalDirectory == null) {
114             return null;
115         }
116         return new File(externalDirectory, FILE_NAME);
117     }
118
119     private boolean isExternalStorageWritable() {
120         return Environment.MEDIA_MOUNTED.equals(Environment.getExternalStorageState());
121     }
122
123     private boolean isExternalStorageReadable() {
124         String state = Environment.getExternalStorageState();
125         return Environment.MEDIA_MOUNTED.equals(state)
126             || Environment.MEDIA_MOUNTED_READ_ONLY.equals(state);
```

```

1 <?xml version="1.0" encoding="utf-8"?>
2 <manifest xmlns:android="http://schemas.android.com/apk/res/android">
3
4     <!-- Required for legacy external storage access on Android 6 through Android 9. -->
5     <uses-permission
6         android:name="android.permission.READ_EXTERNAL_STORAGE"
7         android:maxSdkVersion="32" />
8     <uses-permission
9         android:name="android.permission.WRITE_EXTERNAL_STORAGE"
10        android:maxSdkVersion="28" />
11
12    <application
13        android:allowBackup="true"
14        android:icon="@drawable/ic_launcher_foreground"
15        android:label="@string/app_name"
16        android:roundIcon="@drawable/ic_launcher_foreground"
17        android:supportsRtl="true"
18        android:theme="@style/Theme.SDCardApp">
19        <activity
20            android:name=".MainActivity"
21            android:exported="true">
22            <intent-filter>
23                <action android:name="android.intent.action.MAIN" />
24
25                <category android:name="android.intent.category.LAUNCHER" />
26            </intent-filter>
27        </activity>
28    </application>
29 </manifest>

```

MainActivity.java </> AndroidManifest.xml </> activity_main.xml ×

```

1 <?xml version="1.0" encoding="utf-8"?>
2 <ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
3     android:layout_width="match_parent"
4     android:layout_height="match_parent"
5     android:background="@color/white"
6     android:fillViewport="true">
7
8     <LinearLayout
9         android:layout_width="match_parent"
10        android:layout_height="wrap_content"
11        android:gravity="center_horizontal"
12        android:orientation="vertical"
13        android:paddingStart="24dp"
14        android:paddingTop="36dp"
15        android:paddingEnd="24dp"
16        android:paddingBottom="36dp">
17
18        <TextView
19            android:id="@+id/titleTextView"
20            android:layout_width="match_parent"
21            android:layout_height="wrap_content"
22            android:gravity="center"
23            android:text="@string/title"
24            android:textColor="@color/text_primary"
25            android:textSize="24sp"
26            android:textStyle="bold" />
27
28        <EditText
29            android:id="@+id/inputEditText"
30            android:layout_width="match_parent"

```

```
MainActivity.java  </> AndroidManifest.xml  </> activity_main.xml x
2      <ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
8          <LinearLayout
55          <Button
60              android:background="@drawable/button_secondary_background"
61              android:text="@string/read"
62              android:textAllCaps="false"
63              android:textColor="@color/color_primary"
64              android:textSize="16sp"
65              android:textStyle="bold" />
66
67          <TextView
68              android:id="@+id/resultLabelTextView"
69              android:layout_width="match_parent"
70              android:layout_height="wrap_content"
71              android:layout_marginTop="28dp"
72              android:text="@string/retrieved_data"
73              android:textColor="@color/text_primary"
74              android:textSize="16sp"
75              android:textStyle="bold" />
76
77          <TextView
78              android:id="@+id/resultTextView"
79              android:layout_width="match_parent"
80              android:layout_height="wrap_content"
81              android:layout_marginTop="10dp"
82              android:background="@drawable/result_background"
83              android:minHeight="120dp"
84              android:padding="16dp"
85              android:text="@string/result_placeholder"
86              android:textColor="@color/text_secondary"
```

```
MainActivity.java  </> AndroidManifest.xml  SDCARD.iml x  </> activity_main.xml
1      <?xml version="1.0" encoding="UTF-8"?>
2      <module type="JAVA_MODULE" version="4">
3          <component name="NewModuleRootManager" inherit-compiler-output="true">
4              <exclude-output />
5              <content url="file://$MODULE_DIR$">
6                  <sourceFolder url="file://$MODULE_DIR$/src" isTestSource="false" />
7              </content>
8              <orderEntry type="inheritedJdk" />
9              <orderEntry type="sourceFolder" forTests="false" />
10         </component>
11     </module>
```

```
MainActivity.java  </> AndroidManifest.xml  SDCARD.iml x  </> misc.xml  </> gradle.xml x
1  <?xml version="1.0" encoding="UTF-8"?>
2  <project version="4">
3      <component name="GradleMigrationSettings" migrationVersion="1" />
4      <component name="GradleSettings">
5          <option name="linkedExternalProjectsSettings">
6              <GradleProjectSettings>
7                  <option name="externalProjectPath" value="$PROJECT_DIR$" />
8                  <option name="modules">
9                      <set>
10                         <option value="$PROJECT_DIR$" />
11                         <option value="$PROJECT_DIR$/app" />
12                     </set>
13                 </option>
14             </GradleProjectSettings>
15         </option>
16     </component>
17 </project>
```

```
gradlew  gradlew.bat  local.properties  gradle-wrapper.properties  build.gradle (:app) x v
1  plugins {
2      id "com.android.application"
3  }
4
5  android {
6      namespace "com.example.sdcard"
7      compileSdk 36
8
9      defaultConfig {
10         applicationId "com.example.sdcard"
11         minSdk 24
12         targetSdk 36
13         versionCode 1
14         versionName "1.0"
15     }
16 }
17
```

OUTPUT:

